

Ligjerata 1f

Puna e gazit të mbyllur gjatë procesit adiabatik

$Q = \text{konst.}$ procesi adiabatik

$$dQ = 0$$

$$d\hat{Q} = du + p dv$$

$$p dv = -du$$

$$A = \int p dv$$

$$A = -\int du = -\int_{T_2}^{T_1} C_v dT, \quad C_v = \frac{1}{2} nR$$

$$A = \frac{1}{2} nR (T_2 - T_1)$$

$$TV^{\kappa-1} = \text{konst}$$

$$T_1 V_1^{\kappa-1} = T_2 V_2^{\kappa-1}$$

$$T_2 = T_1 \frac{V_1^{\kappa-1}}{V_2^{\kappa-1}} = T_1 \left(\frac{V_1}{V_2} \right)^{\kappa-1}$$

$$A = \frac{1}{2} nR \left(T_1 \left(\frac{V_1}{V_2} \right)^{\kappa-1} - T_1 \right)$$

$$A = \frac{1}{2} nR T_1 \left[\left(\frac{V_1}{V_2} \right)^{\kappa} - 1 \right]$$

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$$pV^{\kappa} = \text{konst}$$

$$p_1 V_1^{\kappa} = p_2 V_2^{\kappa}$$

$$\left(\frac{V_1}{V_2} \right)^{\kappa} = \frac{p_2}{p_1}$$

$$\frac{V_1}{V_2} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{\kappa}}$$

$$\left(\frac{V_1}{V_2} \right)^{\kappa-1} = \left(\frac{p_2}{p_1} \right)^{\frac{\kappa-1}{\kappa}}$$

$$A = \frac{1}{2} nR T_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{\kappa-1}{\kappa}} - 1 \right]$$